

Network Address: 10.0.0.0/16

Network Name	Host	Host+2	Block size	Host bits	Prefix	Network Address
Sheldon	812	814	1024	10	22	10.0.0.0/22
Leonard	360	362	512	9	23	10.0.4.0/23
Raj	275	277	512	9	23	10.0.6.0/23
Howard	190	192	256	8	24	10.0.8.0/24
Penny	58	60	64	6	26	10.0.9.0/26
Apartment	21	23	32	5	27	10.0.9.64/27

0 0 0 0 0 0 | 0 0

0 0 0 0 0 0 | 0 0

$$000001 \mid 00$$

0	0	0	0	0	1	0	0
---	---	---	---	---	---	---	---

0	0	0	0	0	1	0	0
0	0	0	0	0	1	1	0

0000101000

$$\begin{array}{r} 00000 = 0 \\ 000010 = 0 \end{array}$$

000010 000

00001001

00001001

0000 1001

00001001
00001001

00001001
00001001

0000-0001

00000000 → 10.0.0.0/16

00000000 $\rightarrow$ 10.0.0.0/22 (Sheldon)

00000000 $\rightarrow$ 10.0.4.0/22

000000000 ≥ 10.4.4.0/23 (1 expanded)

00 00000000 → 10.0.6.0/23 (Pc: 2)

[illegible]

0000000000 → 10.0.8.0/22

0000000000 $\rightarrow$ 10.0.8.0/23

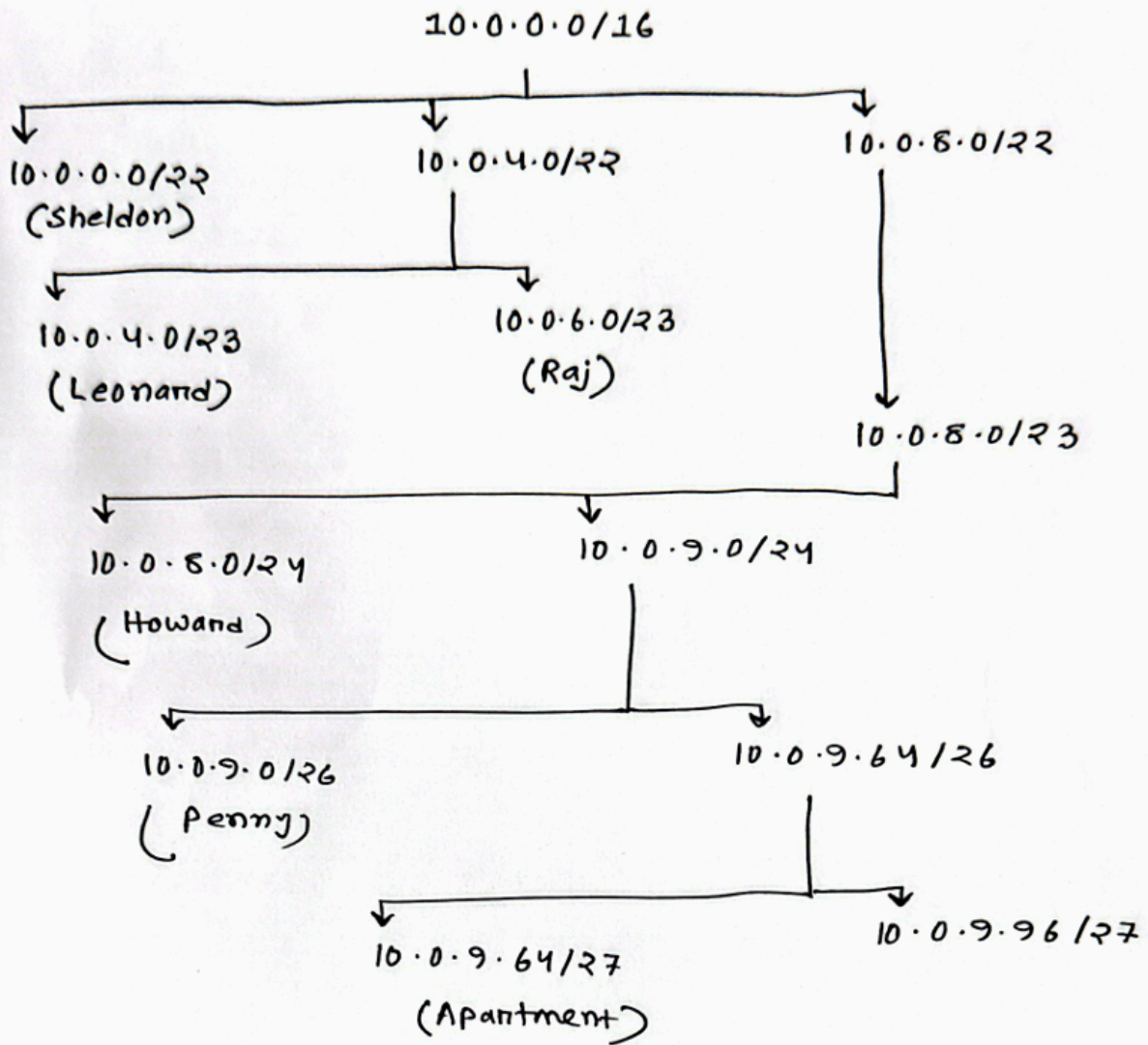
0000 00000 $\rightarrow 10.0.8.0/24$
0000 00000 $\rightarrow 10.0.8.0/24$

10.0.9.0/24

00000000 → 10.0.9.0/26

01|000000 → 10.0.9.64 / 26

01 0 00000 $\rightarrow$ 10.0.9.64/27
01 1 00000 $\rightarrow$.



IP Address Table

Network Zone	Subnet Address / Mask	Default Gateway	Servers (IP)	Example End Devices	Router Interface
Theoretical Physics Lab (Sheldon)	10.0.0.0/22 (255.255.252.0)	10.0.0.1	DHCP: 10.0.0.2 Email: 10.0.0.20	PC1: 10.0.0.10 PC2: 10.0.0.11	Router-A G0/0 → Sheldon LAN
Experimental Physics Lab (Leonard)	10.0.4.0/23 (255.255.254.0)	10.0.4.1	DHCP: 10.0.4.2	PC1: 10.0.4.10 PC2: 10.0.4.11	Router-B G0/0 → Leonard LAN
Particle Astrophysics Center (Raj)	10.0.6.0/23 (255.255.254.0)	10.0.6.1	DHCP: 10.0.6.2 Email: 10.0.6.20	PC1: 10.0.6.10 PC2: 10.0.6.11	Router-C G0/0 → Raj LAN
JPL Engineering Bay (Howard)	10.0.8.0/24 (255.255.255.0)	10.0.8.1	DHCP: 10.0.8.2	PC1: 10.0.8.10 PC2: 10.0.8.11	Router-B G0/1 → Howard LAN

Cheesecake Factory Back Office (Penny)	10.0.9.0/26 (255.255.255.192)	10.0.9.1	DHCP: 10.0.9.2	PC1: 10.0.9.10 PC2: 10.0.9.11	Router-A G0/1 → Penny LAN
Pasadena Apartment Committee	10.0.9.64/27 (255.255.255.224)	10.0.9.65	Web: 10.0.9.70 DNS: 10.0.9.71 Support: 10.0.9.72	PC1: 10.0.9.66 PC2: 10.0.9.67	Router-D G0/0 → Apartment LAN

Router A (interfaces)

```

enable
configure terminal
interface g0/0
ip address 10.0.0.1 255.255.252.0
no shutdown
exit
interface g0/1
ip address 10.0.9.1 255.255.255.192
no shutdown
exit
interface s0/0/0
ip address 192.168.1.1 255.255.255.252
no shutdown
exit

```

Router B (interfaces)

```

enable

```

```
configure terminal
interface g0/0
ip address 10.0.4.1 255.255.254.0
no shutdown
exit
interface g0/1
ip address 10.0.8.1 255.255.255.0
no shutdown
exit
interface s0/0/0
ip address 192.168.1.2 255.255.255.252
no shutdown
Exit
interface s0/0/1
ip address 192.168.2.1 255.255.255.252
no shutdown
exit
```

Router C (interfaces)

```
enable
configure terminal
interface g0/0
ip address 10.0.6.1 255.255.254.0
no shutdown
exit
interface g0/1
ip address 10.0.9.65 255.255.255.224
no shutdown
exit
interface s0/0/0
ip address 192.168.2.2 255.255.255.252
no shutdown
exit
```

Dynamic Routing (B router)

```
router rip
version 2
no auto-summary
network 10.0.4.0
network 10.0.8.0
network 192.168.2.0
exit
```

Static Routing (Router A)

configure terminal

```
ip route 10.0.4.0 255.255.254.0 192.168.1.2
```

```
ip route 10.0.6.0 255.255.254.0 192.168.1.2
```

```
ip route 10.0.9.64 255.255.255.224 192.168.1.2
```

Static Routing (Router c)

configure terminal

```
ip route 10.0.0.0 255.255.252.0 192.168.2.1
```

```
ip route 10.0.9.0 255.255.255.192 192.168.2.1
```

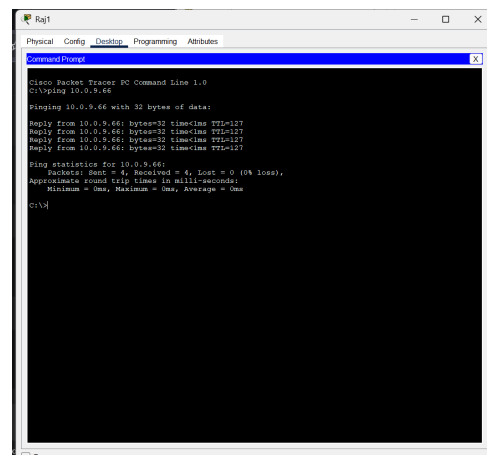
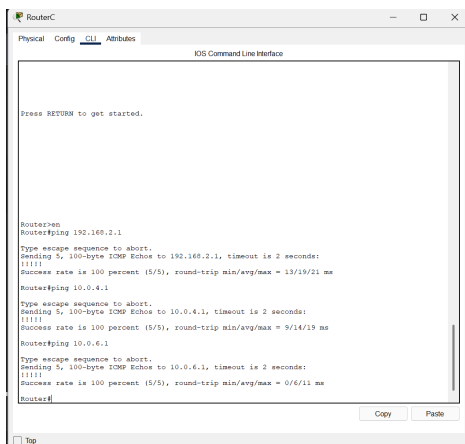
Floating Static Route (Router C)

configure terminal

```
ip route 10.0.6.0 255.255.254.0 192.168.2.1 25
```

end

write memory



1. Apartment Committee Network – Static IP Configuration

Subnet: 10.0.9.64 /27

Subnet Mask: 255.255.255.224

Default Gateway: 10.0.9.65

Apartment PC-1:

IP Address:10.0.9.66, Gateway:10.0.9.65 ,DNS:10.0.9.71

Apartment PC-2:

IP Address:10.0.9.67, Gateway:10.0.9.65 ,DNS:10.0.9.71

2. Apartment Servers

Web Server:

IP Address: 10.0.9.70

Subnet Mask: 255.255.255.224

Default Gateway: 10.0.9.65

DNS Server: 10.0.9.71

DNS Server:

IP Address: 10.0.9.71

Subnet Mask: 255.255.255.224

Default Gateway: 10.0.9.65

DNS Server: 10.0.9.71

CaltechSupport Server:

IP Address: 10.0.9.72

Subnet Mask: 255.255.255.224

Default Gateway: 10.0.9.65

DNS Server: 10.0.9.71

3. Web Server Configuration

Edit index.html = “**Bazinga! We’ve got your back!**”

4. DNS Server Configuration:

Domain Name: caltechsupport.com

Record Type: A Record

IP Address: 10.0.9.72

5. DHCP Server Configuration

DHCP Server Static IP:

IP Address: 10.0.0.2

Subnet Mask: 255.255.252.0

Default Gateway: 10.0.0.1

DNS Server: 10.0.9.71

DHCP Pool Configuration:

Pool Name: SHELDON_DHCP

Network: 10.0.0.0

Subnet Mask: 255.255.252.0

Default Gateway: 10.0.0.1

DNS Server: 10.0.9.71

Start IP: 10.0.0.50

5.2 Leonard –

DHCP Server Static IP

IP Address: 10.0.4.2

Subnet Mask: 255.255.254.0

Default Gateway: 10.0.4.1

DNS Server: 10.0.9.71

DHCP Pool

Pool Name: LEONARD_DHCP

Network: 10.0.4.0

Subnet Mask: 255.255.254.0

Default Gateway: 10.0.4.1

DNS Server: 10.0.9.71

Start IP: 10.0.4.50

5.3 Raj

DHCP Server Static IP

IP Address: 10.0.6.2

Subnet Mask: 255.255.254.0

Default Gateway: 10.0.6.1

DNS Server: 10.0.9.71

DHCP Pool

Pool Name: RAJ_DHCP

Network: 10.0.6.0

Subnet Mask: 255.255.254.0

Default Gateway: 10.0.6.1

DNS Server: 10.0.9.71

Start IP: 10.0.6.50

5.4 Howard

DHCP Server Static IP

IP Address: 10.0.8.2

Subnet Mask: 255.255.255.0

Default Gateway: 10.0.8.1

DNS Server: 10.0.9.71

DHCP Pool

Pool Name: HOWARD_DHCP

Network: 10.0.8.0

Subnet Mask: 255.255.255.0

Default Gateway: 10.0.8.1

DNS Server: 10.0.9.71

Start IP: 10.0.8.50

5.5 Penny

DHCP Server Static IP

IP Address: 10.0.9.2

Subnet Mask: 255.255.255.192

Default Gateway: 10.0.9.1

DNS Server: 10.0.9.71

DHCP Pool

Pool Name: PENNY_DHCP

Network: 10.0.9.0

Subnet Mask: 255.255.255.192

Default Gateway: 10.0.9.1

DNS Server: 10.0.9.71

Start IP: 10.0.9.10

6. Email Server Configuration (Sheldon ↔ Raj)

Sheldon Email Server

IP Address: 10.0.0.20

SMTP: ON

POP3: ON

User: sheldon

Password: 1234

Domain: tpl.caltech.edu

Raj Email Server

IP Address: 10.0.6.20

SMTP: ON

POP3: ON

User: raj

Password: 1234

Domain: pac.caltech.edu

7. Email Client Configuration

Sheldon PC

Email Address: sheldon@tpl.caltech.edu

Incoming Mail Server: 10.0.0.20

Outgoing Mail Server: 10.0.0.20

Username: sheldon

Password: 1234

Raj PC

Email Address: raj@pac.caltech.edu

Incoming Mail Server: 10.0.6.20

Outgoing Mail Server: 10.0.6.20

Username: raj

Password: 1234

8. Functional Testing Commands

DNS Reachability

ping 10.0.9.71

Web Server Reachability

ping 10.0.9.72

Web Access Test

`http://caltechsupport.com`

Inter-Network Connectivity Example

`ping 10.0.6.50`